

No naive Hecke iso $B_{\ell+1}^{(\lambda)} V_\lambda \cong V_{\lambda^{(\geq 2)}}:$ a precise computational refutation

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Abstract

The May-13 closed-form theorem says $\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}}$ in the stable regime, where $\lambda^{(\geq 2)}$ is the decoration shape (column 1 deleted). The natural next question is whether this is a *representation-theoretic* isomorphism $B_{\ell+1}^{(\lambda)} V_\lambda \cong V_{\lambda^{(\geq 2)}}$ as $H_q(S_{n-\ell})$ -modules under some embedding of $H_q(S_{n-\ell})$ into $H_q(S_n)$.

We refute this in its most natural form. For the stable test case $\lambda = (3, 2, 1, 1, 1)$ at $n = 8$, where $\dim B = 2 = f^{(2,1)}$ and $\dim V_{\lambda^{(\geq 2)}} = \dim V_{(2,1)} = 2$, direct computation in the Hoefsmit seminormal basis shows that the only Hecke generators T_j ($1 \leq j \leq n-1$) that preserve B are

$$T_1 \quad (\text{acting as scalar } -1) \quad \text{and} \quad T_\ell = T_5 \quad (\text{acting as scalar } q).$$

All other T_j fail to preserve B (they extend B to a strictly larger subspace of V_λ). Moreover, no product $T_i T_j$ or braid triple $T_i T_j T_i$ preserves B unless it is already in $\langle T_1, T_5 \rangle$. Hence no Hecke subalgebra generated by simple T_j 's or their short products carries a non-trivial $H_q(S_{n-\ell})$ -action on B .

This forces one of three resolutions:

- (a) the iso, if it exists, is via *exotic* operators in $H_q(S_n)$ (e.g. Murphy/Jucys elements or specific intertwiners) — none of which were located in this session;
- (b) the iso holds only at the level of *vector spaces* (the chain bijection $\text{Chain}(\lambda) \leftrightarrow \text{SYT}(\lambda^{(\geq 2)})$, established in the May-13 paper) and has no Hecke-module upgrade;
- (c) the iso holds via a *subquotient* or *induced* Hecke structure that is not a literal subalgebra of $H_q(S_n)$.

We expand on (a)–(c) and the structural surprises uncovered along the way.

1 Setup

We retain the notation of the May-13 paper (2026-05-13-multi-corner-dimension.tex). In particular: $H_q(S_n)$ is the Iwahori–Hecke algebra with generators $T_1, \dots, T_{n-1}$; V_λ is the Specht module in the Hoefsmit seminormal basis; $R'_j := (T_{j-1} + 1) \cdots (T_1 + 1)$ and

$$B_{\ell+1}^{(\lambda)} V_\lambda := R'_{\ell+1} R'_{\ell+2} \cdots R'_n V_\lambda \subseteq V_\lambda,$$

the iterated-image subspace at the sharp index $\ell+1$. The decoration shape is $\lambda^{(\geq 2)} := (\lambda_1 - 1, \lambda_2 - 1, \dots)$ (drop zero parts).

The May-13 paper proves (modulo the technical ingredients there) that in the stable regime,

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}}.$$

Since $|\lambda^{(\geq 2)}| = n - \ell$, the dimension matches that of the Specht module $V_{\lambda^{(\geq 2)}}$, which is irreducible over $H_q(S_{n-\ell})$ at generic q .

Question 1 (Target). Is there a copy of $H_q(S_{n-\ell})$ inside $H_q(S_n)$ acting on $B_{\ell+1}^{(\lambda)} V_\lambda$ such that the resulting module is isomorphic to $V_{\lambda^{(\geq 2)}}$?

The most natural candidates for such a copy are:

- (A) the *standard parabolic* embedding $H_q(S_{n-\ell}) = \langle T_{\ell+1}, T_{\ell+2}, \dots, T_{n-1} \rangle$ acting on positions $\{\ell + 1, \dots, n\}$;
- (B) the *shifted parabolic* embedding $H_q(S_{n-\ell}) = \langle T_2, T_3, \dots, T_{n-\ell} \rangle$ acting on positions $\{2, 3, \dots, n - \ell + 1\}$ (or any other run);
- (C) a copy generated by Jucys–Murphy elements, intertwiners, or braid words.

Below we eliminate (A) and (B) outright by direct computation, then discuss (C).

2 Computational refutation

We work modulo $P = 100\,003$ at $q = 11$ in the Hoefsmit seminormal matrix model (script `compute_dim_B.py`, May-13). The mod- P rank equals the rank over $\mathbb{Q}(q)$ for any non-special q .

2.1 The test case $\lambda = (3, 2, 1, 1, 1)$ at $n = 8$

Here $\ell = 5$, $\dim V_\lambda = 64$, and $\dim B_6^{(\lambda)} V_\lambda = 2 = f^{(2,1)}$ (where $\lambda^{(\geq 2)} = (2, 1)$).

Theorem 2 (Direct refutation). *For each $j \in \{1, 2, \dots, 7\}$, computation gives:*

j	T_j preserves B ?	action / failure mode
1	yes	scalar -1
2	<i>no</i>	$\dim(B + T_2 B) = 4$
3	<i>no</i>	$\dim(B + T_3 B) = 4$
4	<i>no</i>	$\dim(B + T_4 B) = 4$
5	yes	scalar q
6	<i>no</i>	$\dim(B + T_6 B) = 4$
7	<i>no</i>	$\dim(B + T_7 B) = 4$

Hence the only generators stabilising B are T_1 and T_5 , both acting as scalars; the subalgebra they generate is the commutative algebra spanned by $1, T_1, T_5, T_1 T_5$, isomorphic to the group algebra $\mathbb{Z}[\zeta_2^2]$ of a Klein 4-group (after rescaling).

Proof (computational). Let V_λ have its Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$, and build T_j as a 64×64 mod- P matrix. Build $\mathcal{B} := R'_6 R'_7 R'_8$ as a matrix product. Apply Gaussian elimination to find a basis W of $\text{Im}(\mathcal{B})$ (two columns).

For each j , form the 64×2 matrix $T_j W$ and compute the rank of $[W | T_j W]$ (mod P). If this rank equals 2, then $T_j W \subseteq \text{span}(W)$ (so T_j preserves B); else it strictly extends B . The table above is the result of this test for each j .

For $j \in \{1, 5\}$, the resulting 2×2 action matrix on W is a scalar ± 1 resp. q , verified by direct comparison to $\pm I_2$ resp. qI_2 . □ □

Corollary 3 (Standard parabolic embedding fails). *The standard $H_q(S_{n-\ell}) = \langle T_{\ell+1}, \dots, T_{n-1} \rangle = \langle T_6, T_7 \rangle$ does not preserve B (neither T_6 nor T_7 does individually). Hence (A) is impossible.*

Corollary 4 (Shifted parabolic embeddings also fail). *For any run $\langle T_a, T_{a+1}, \dots, T_{a+n-\ell-2} \rangle$ of $n - \ell - 1 = 2$ consecutive generators (i.e. $a \in \{1, 2, \dots, 6\}$ with $a + 2 \leq 8$), at least one generator in the run fails to preserve B , since T_2, T_3, T_4, T_6, T_7 all fail to preserve B .*

(Only T_1, T_5 work, and they are non-consecutive, so they don't generate a Hecke algebra of type A.)

2.2 No short-word extensions help

Proposition 5 (Short-word search). *For each ordered pair (i, j) with $1 \leq i, j \leq 7$, the product $T_i T_j$ preserves B if and only if both T_i and T_j already preserve B individually (i.e. $i, j \in \{1, 5\}$). Similarly for braid triples $T_i T_j T_i$ with $|i - j| = 1$: none of them preserve B (other than via individual factors, which doesn't apply here).*

Proof (computational). Enumerate all 49 products $T_i T_j$ and all ~ 12 braid triples $T_i T_j T_i$ with $|i - j| \leq 1$, $i \neq j$; for each, apply to the 2-dim basis W of B and check whether the result lies in W . Output: only products with $i, j \in \{1, 5\}$ preserve B . Braid triples (all involving at least one of T_2, T_3, T_4, T_6, T_7) all fail. $\square$ $\square$

2.3 Multiple test cases confirm the pattern

The same phenomenon — only $T_1, \dots, T_k$ (acting as scalar -1 , for some small k) and T_ℓ (acting as scalar q) preserve B — holds across multiple stable test cases:

λ	n	ℓ	$T_j = -1$ stable	$T_j = q$ stable
$(3, 2, 1, 1, 1)$	8	5	$\{1\}$	$\{5\}$
$(3, 2, 1, 1, 1, 1)$	9	6	$\{1, 2\}$	$\{6\}$
$(4, 2, 1, 1, 1, 1)$	10	6	$\{1\}$	$\{6\}$
$(3, 3, 1, 1, 1, 1)$	10	6	$\{1\}$	$\{6\}$
$(4, 3, 1, 1, 1)$	10	5	$\emptyset$	$\{5\}$

In every case, the stable generators are $\{T_1, \dots, T_k\} \cup \{T_\ell\}$ for some $k = k(\lambda) \in \{0, 1, 2\}$, and they all act as scalars. The threshold k depends on λ in a non-obvious way — it is not just $\ell - r - 1$ or $q - 2$; nor is it just the rank-zero predictor.

3 Why the naive iso fails

3.1 Comparison to the Littlewood–Richardson decomposition

A natural place where one would expect to find $V_{\lambda(\geq 2)}$ inside V_λ is the $V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)}$ -isotypic component of $V_\lambda \downarrow_{H_q(S_\ell) \times H_q(S_{n-\ell})}$.

By Littlewood–Richardson, this isotypic component has multiplicity $c_{(1^\ell), \lambda(\geq 2)}^\lambda$. For our λ , the skew shape $\lambda/(1^\ell)$ has one cell in each row of λ (since each row contributes column 1 to (1^ℓ)); this is a vertical strip iff each row has ≤ 1 cell removed, which is automatic, and the LR condition uniquely identifies $\lambda^{(\geq 2)}$ as the partition obtained by deleting the strip. Thus $c_{(1^\ell), \lambda(\geq 2)}^\lambda = 1$, and the isotypic component has dimension exactly $f^{\lambda(\geq 2)}$ — matching $\dim B$.

Observation 6 (The fatal mismatch). The $V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)}$ -isotypic component of $V_\lambda \downarrow_{H_q(S_\ell) \times H_q(S_{n-\ell})}$ is *not* B . The isotypic component requires $T_j v = -v$ for all $j \in \{1, \dots, \ell-1\}$ (the sign rep on the first factor), but Theorem 2 shows that T_2, T_3, T_4 *do not even preserve* B .

So there are two distinct $f^{\lambda(\geq 2)}$ -dimensional subspaces of V_λ at the relevant size:

- the LR isotypic component $V_{(1^\ell)} \otimes V_{\lambda(\geq 2)} \subseteq V_\lambda$ (which carries an honest $H_q(S_{n-\ell})$ -action via $T_{\ell+1}, \dots, T_{n-1}$, and an $H_q(S_\ell)$ -action as the sign rep);
- the iterated-image $B = B_{\ell+1}^{(\lambda)} V_\lambda$ (which carries only the scalar action of $T_1, \dots, T_k, T_\ell$).

These have the same dimension by an LR-style coincidence, but they are genuinely different subspaces, and the second has a fundamentally weaker Hecke-action structure than the first.

3.2 Hoefsmit-support structure of B

Inspecting the Hoefsmit-basis decomposition of vectors in B for $\lambda = (3, 2, 1, 1, 1)$ reveals that the support is broad and mixed: a typical basis vector of B has nonzero coefficients on $\sim 20+$ SYTs of λ , including SYTs in which the column-1 entries are *not* a prefix of $\{1, 2, \dots, \ell\}$. For example,

$$T_0 = [(1, 3, 4) (2, 5) (6) (7) (8)]$$

appears in the support: entry 3 is at $(1, 2)$ (not in column 1), entry 5 at $(2, 2)$, and entries $\{6, 7, 8\}$ in the bottom three column-1 rows. Thus the column-1 spine carries $\{1, 2, 6, 7, 8\}$, not $\{1, 2, 3, 4, 5\}$, demonstrating that B is *not* spanned by SYTs with column 1 filled by the smallest entries.

This already implies that B is not the $V_{(1^\ell)}$ -isotypic component (which *is* spanned by SYTs with column 1 filled by $\{1, \dots, \ell\}$).

4 Where might the iso live? The three resolutions

The numerical coincidence $\dim B = f^{\lambda(\geq 2)}$ is real (May-13 closed-form theorem, verified across 9 shapes in the stable regime). What it does *not* naturally upgrade to is a Hecke-equivariant iso with $V_{\lambda(\geq 2)}$ under any of the obvious embeddings (A) or (B). The remaining possibilities:

(a) Exotic operators in $H_q(S_n)$

There might exist $X_1, \dots, X_{n-\ell-1} \in H_q(S_n)$, not of the form T_i or simple products, that preserve B and satisfy the Hecke relations of $H_q(S_{n-\ell})$. A candidate source is the family of *Jucys–Murphy elements* $L_k = T_{k-1} T_{k-2} \cdots T_1 \cdot T_1 \cdots T_{k-1}$ together with *intertwining operators* Φ_k that exchange axial distances. Neither was located in this session; an exhaustive search would require enumeration of much longer words.

(b) Vector-space-only iso

The chain bijection $\text{Chain}(\lambda) \leftrightarrow \text{SYT}(\lambda^{(\geq 2)})$ (Section 3 of the May-13 paper) is a genuine bijection between two combinatorial sets and induces an explicit isomorphism of *vector spaces* $\mathbb{C}^{\text{Chain}(\lambda)} \cong V_{\lambda(\geq 2)}$. This bijection, however, is purely combinatorial; it does not on its own give a Hecke action on B .

The honest content of the May-13 paper is:

$$\dim B = \# \text{Chain}(\lambda) = f^{\lambda(\geq 2)},$$

not an iso of representations.

(c) Subquotient / induced structure

B might be a sub-quotient of V_λ as an $H_q(S_n)$ -module whose graded pieces include $V_{\lambda(\geq 2)}$ (now an $H_q(S_n)$ -module via some restriction-of-scalars or pullback). Or B might be the image of an H_q -equivariant morphism from an induced module $\text{Ind}_{H_q(S_\ell) \times H_q(S_{n-\ell})}^{H_q(S_n)} V_{(1^\ell)} \boxtimes V_{\lambda(\geq 2)}$. Both possibilities are consistent with the data but require representation-theoretic machinery not developed here.

5 Computational evidence: stability threshold (backup work)

While probing the iso theorem, we ran the suggested backup target (PROVE.MD, Section “Backup target”): the stability threshold. The result is striking and *contradicts* the May-13 paper’s empirical claim that “ $q \geq 2$ trailing 1-rows suffices.”

Theorem 7 (Stability threshold is shape-dependent and non-monotone). *Let $\hat{\lambda}$ be a multi-column base partition (no trailing 1-rows) and write $\lambda(\hat{\lambda}, q) := (\hat{\lambda}, 1^q)$. The minimum q for which $\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda(\geq 2)}$ holds is not a universal constant; it depends on $\hat{\lambda}$. Moreover, the matching set $\{q : \dim B = f^{\lambda(\geq 2)}\}$ is not an up-set in q in general.*

Proof (computational table). Direct mod- P computation gives the following table, where $\dim B$ is computed via `compute_dim.B.py` mod $P = 100\,003$ at $q = 11$:

$\hat{\lambda} \setminus q$	0	1	2	3	4	5
(3, 2), target $f^{(2,1)} = 2$	2✓	2✓	2✓	2✓	2✓	2✓
(4, 2), target $f^{(3,1)} = 3$	3✓	6 ×	3✓	3✓	3✓	—
(3, 3), target $f^{(2,2)} = 2$	1 ×	3 ×	2✓	2✓	2✓	—
(4, 3), target $f^{(3,2)} = 5$	4 ×	7 ×	12 ×	5✓	—	—
(5, 2), target $f^{(4,1)} = 4$	5 ×	8 ×	12 ×	—	—	—
(3, 2, 2), target $f^{(2,1,1)} = 3$	3✓	3✓	3✓	—	—	—

Entries marked ✓: $\dim B = f^{\hat{\lambda}(\geq 2)}$. Entries marked ×: failure, with the actual $\dim B$ in bold. □
□

Remark 8 (Non-monotonicity at (4, 2)). The (4, 2) family shows the matching set $\{q : \dim B = f^{\lambda(\geq 2)}\} = \{0, 2, 3, 4\}$ — *not an up-set*: $q = 0$ matches, $q = 1$ fails ($\dim = 6 \neq 3$), then $q \geq 2$ matches again. This refutes any monotone-in- q stability hypothesis.

Remark 9 (Refining the May-13 threshold claim). The May-13 paper conjectured “ $q \geq 2$ suffices empirically.” This is *wrong*: for $\hat{\lambda} = (4, 3)$, the threshold is $q \geq 3$ (the $q = 2$ case fails with $\dim = 12 \neq 5$); for $\hat{\lambda} = (5, 2)$, we have failure all the way through $q = 2$ and the threshold is ≥ 3 (untested at $q \geq 3$).

The threshold should be conjectured shape-dependently. A first guess:

$$q_0(\hat{\lambda}) \stackrel{?}{=} \ell(\hat{\lambda}) + (\text{depth of recursion}),$$

but this is heuristic and unverified.

Remark 10 (Failure dimensions also non-obvious). At the non-matching cases, the actual $\dim B$ does not seem to match a simple alternative formula. For example, $(4, 2)$ at $q = 1$ gives $\dim = 6 = 2 \cdot 3 = 2f^{(3,1)}$, and $(4, 3)$ at $q = 2$ gives $\dim = 12$ versus prediction 5 (no obvious factor). The structure of B in the non-stable regime remains open.

Remark 11 (Strict stable regime exhibits the formula). When q exceeds the shape-dependent threshold $q_0(\widehat{\lambda})$, the formula $\dim B = f^{\lambda(\geq 2)}$ holds and is robust — this is consistent with the May-13 closed-form theorem in the stable regime, just with a stricter (and correct) definition of stability.

6 What was learned

1. The dim equality $\dim B = f^{\lambda(\geq 2)}$ is real (MAY-13 paper, modulo Phase A generalisation) *in the strict stable regime*, but the strict regime is shape-dependent and narrower than the May-13 paper estimated.
2. The naive Hecke-iso upgrade is *false*: there is no run $\langle T_a, T_{a+1}, \dots \rangle$ of $n - \ell - 1$ consecutive generators preserving B .
3. The May-13 strategy note (PROVE.MD Step 1) suggested that “ B is fixed under $(T_{j-1} + 1)$ for $j > \ell$ ” — this is consistent only in the weak sense that T_ℓ acts as scalar q on B ($B \subseteq E_\ell^+$). Higher T_j ($j > \ell$) do *not* preserve B as a subspace.
4. The number $k(\lambda)$ such that $T_1, \dots, T_k$ act as -1 on B is shape-dependent and not simply expressible in ℓ, r, q alone.
5. The stability threshold $q_0(\widehat{\lambda})$ is shape-dependent and the matching set $\{q : \dim B = f^{\lambda(\geq 2)}\}$ can be non-monotone (e.g. $(4, 2)$: matches at $q = 0$, fails at $q = 1$, matches at $q \geq 2$). This is a substantive refinement of the May-13 paper’s claim.
6. The iso, if it exists, requires non-obvious operators (option (a)) or a subquotient / induced structure (option (c)). Locating those is the next concrete research target.

7 Honest status

The target theorem of PROVE.MD (Section “The target theorem”) is *not* proved here; in its naive form (any obvious parabolic embedding of $H_q(S_{n-\ell})$) it is *refuted*. The dim-formula program is robust on its own — the May-13 paper, with its closed-form SYT count in the stable regime, is the strongest statement we currently have. The rep-theoretic content remains conjectural and now sits as an open question of structural type, not a numerical one.